Covariant flux and physical-domain limits of regular de Sitter shells

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Abstract

A positive self-adjoint radial operator does not by itself identify the physical perturbation space at a characteristic cone. For a registered doubled Einstein–scalar de Sitter shell, we derive the covariant presymplectic current on its generic longitudinal scalar solution slice. Its radial component is proportional to the reduced Green current at a common spectral value, but has a different mixed-spectral expression for general pairs and packets. We verify the even shell cancellation and the completed moving-shell current in the odd sector. Under explicit uniform characteristic asymptotics for actual solutions, the cone limit is a generally nonzero scalar null symplectic bilinear. This conditional statement is distinguished from an exact smooth finite-energy test-scalar example in a flat cone. The known singularity of a normalizable tensor zero mode is recovered componentwise, while cutoff estimates retain that mode in the energy completion. Smooth residual translations have no shrinking-vertex charge, without implying zero charge at finite null cuts. These calculations refine the scope of the existing radial theorem; they neither construct global radiative solutions of the registered shell problem nor establish physical completeness or nonlinear stability.

Status. Technical continuation draft; not an APS submission. The earlier spectral capsule [1] is an immutable input, not a new edition of this note.

1. Problem, model and inherited result

The regular-shell checkpoint [1] proves positive scalar spectral bounds and the tensor mass threshold for specified radial realizations. Its next physical question is which covariant perturbations, boundary data and gauge quotient those realizations describe. Three distinctions are essential: smooth configurations versus their finite-energy completion; a shrinking vertex limit versus a finite segment of the null cone; and an antisymmetric symplectic flux versus a positive quadratic energy. We keep all three explicit. Our main result is the local current and its shell balance for the generic scalar sector of the registered model.

Take signature $(-++++)$, $8\pi G_5 = 1$, and two identical interiors with one common shell. The action, including its variational boundary completion, is

$$S = \sum_{i=\pm} \left\{ \int_{M_i} \epsilon_g \left[\frac{R}{2} - \frac{1}{2}(\nabla\phi)^2 - U(\phi) \right] + \int_\Sigma \epsilon_h K_i \right\} - \int_\Sigma \epsilon_h \sigma(\phi). \tag{1}$$

Both shell normals point toward increasing interior radius. With $\eta = \phi - 1$, the registered functions are

$$W = \frac{1}{3} + \eta^2 + \frac{\eta^3}{3}, \qquad U = \frac{1}{2}W_{,\eta}^2 - \frac{2}{3}W^2, \qquad \sigma = 2W + \delta(1 + c + c\eta), \quad c = \frac{2}{1.0357712571566784} - \frac{4}{3}. \tag{2}$$

The decimal denotes an exact rational. The existing regular branch covers $0 < \delta \le 0.002$ and $|\eta| \le 1/100$. We do not recertify that branch. Write

$$g = dy^2 + \rho(y)^2 \gamma_{\mu\nu} dx^\mu dx^\nu, \quad \mathrm{Ric}(\gamma) = 3\gamma, \quad 0 < y \le T_b, \quad a = \rho'/\rho, \quad p = \phi'.$$

The equations and background junctions are

$$p' = U_{,\phi} - 4ap, \quad a' = -\rho^{-2} - p^2/3, \quad a^2 = \rho^{-2} + p^2/12 - U/6, \qquad a_b = \sigma_b/6, \quad p_b = -\sigma_{,\phi,b}/2. \tag{3}$$

At a fixed background, $p < 0$ on $(0, T_b]$, $a_b > 0$, and $B_b = (p'/p + \sigma_{,\phi\phi}/2)_b > 0$. The regular expansions have $\rho = y + O(y^3)$ and $p = \kappa y + O(y^3)$, $\kappa \neq 0$, with the differentiated remainders used below. Smooth geometric extension statements use coefficient functions smooth in y^2, as in the analytic regular branch.

For orientation, the inherited generic master coefficients are

$$P = \rho^2 \Psi, \quad \alpha = (\rho^2 p^2)^{-1}, \quad q = \frac{2}{3\rho^2}, \quad w = (\rho^4 p^2)^{-1}, \quad -(\alpha P')' + qP = \Lambda w P, \quad \mu^2 = \Lambda - 4. \tag{4}$$

The even endpoint condition is $\alpha_b P_b' = \Lambda P_b/\beta$ with $\beta = \rho_b^4 p_b^2 B_b > 0$; the odd one is $\alpha_b P_b' = -c_o P_b$, $c_o = (3a_b\rho_b^2)^{-1} > 0$. The even Hilbert norm includes $|P_b|^2/\beta$, whereas the odd norm contains only $\int w|P|^2 dy$. The checkpoint establishes a strictly positive scalar μ^2 lower bound at each fixed detuning, essential threshold $9/4$, and the even tensor spectral set $\{0\} \cup [9/4, \infty)$, with $[9/4, \infty)$ for odd tensors. These remain statements about their specified operators. No uniform positive physical mass floor follows as $H_b = 1/\rho_b \to 0$.

2. Action-derived current on the generic scalar slice

For $h_{AB} = \delta g_{AB}$, $h = g^{AB} h_{AB}$ and $\chi = \delta\phi$, fix the bulk Lee–Wald potential by

$$\theta^A = \sqrt{-g}\left[\frac{1}{2}(\nabla_B h^{AB} - \nabla^A h) - \nabla^A \phi\, \chi\right], \qquad \omega^A(1,2) = \delta_1\theta^A(2) - \delta_2\theta^A(1). \tag{5}$$

Here h^{AB} raises the covariant variation, rather than denoting δg^{AB}. Variations commute. The displayed currents are coordinate vector densities. In particular ω^y points toward increasing y; the outward flux from $\epsilon < y < T_b$ at its lower boundary is $-\omega^y$. Complex solutions may be paired by conjugating one argument after complexification.

The formalism and its boundary ambiguities are established [2–5]. Adding a boundary Lagrangian changes the potential by a field variation; a spacetime exact improvement changes the current by a corner term. Equation (5), the two GHY terms, and the one shell action specify our convention. We add no cone counterterm and do not infer a null-boundary phase space from (5).

An explicit derivation uses the commuting nonlinear family $g_{yy} = e^{2n}$, $g_{y\mu} = 0$, $g_{\mu\nu} = \rho^2 e^{2\psi}\gamma_{\mu\nu}$, $\phi = \phi_0 + X$. This family evaluates the second variation; it is not assumed to satisfy nonlinear equations. With $K = 4(a + \psi')e^{-n}$, direct contraction in (5) gives

$$\theta^y + \delta(\sqrt{-h}K) = 3\sqrt{-h}K\,\delta\psi - \sqrt{-h}e^{-n}\phi'\delta\phi = \pi^{\mu\nu}\delta h_{\mu\nu} + \pi_\phi \delta\phi, \tag{6}$$

where $\pi^{\mu\nu} = \sqrt{-h}(Kh^{\mu\nu} - K^{\mu\nu})/2$ and $\pi_\phi = -\sqrt{-h}e^{-n}\phi'$. The field-space curl of the GHY variation vanishes. Thus the radial canonical and covariant currents agree pointwise on this zero-shift slice. General variations with shift still require their ordinary tangential corner term.

In the generic scalar subspace the longitudinal constraints give

$$N = -2\Psi, \qquad \Psi' = -2a\Psi - pX/3, \qquad pX' = p'X + 3\rho^{-2}(\Box_\gamma + 4)\Psi - 2p^2\Psi. \tag{7}$$

Here N is the linear lapse perturbation. These implications require independent scalar gradient and trace-free Hessian tensors. The constant and Hessian-degenerate harmonic subspaces require separate treatment; entire mass values are not excluded.

For a direct check, restrict the nonlinear family to $n = -2\psi$. Factoring out $\sqrt{-\gamma}$, its potentials are

$$\theta^y = -\rho^4 e^{6\psi}[4\delta\psi' + 12(a+\psi')\delta\psi + (p+X')\delta X], \quad \theta^\mu = -\rho^2[D^\mu\delta\psi + 6\delta\psi D^\mu\psi + D^\mu X\delta X].$$

Their antisymmetric variations give a radial gravity term $-12\rho^4(\Psi_1\Psi_2' - \Psi_2\Psi_1')$ and a scalar–metric term $-6\rho^4 p(\Psi_1 X_2 - \Psi_2 X_1)$. Applying the momentum constraint in (7) combines them into the following result, per interior:

$$\boxed{\omega^y = \sqrt{-\gamma}\rho^4[X_1X_2' - X_2X_1' - 2p(\Psi_1X_2 - \Psi_2X_1)]}, \tag{8}$$

$$\boxed{\omega^\mu = \sqrt{-\gamma}\rho^2[X_1D^\mu X_2 - X_2D^\mu X_1 + 6(\Psi_1D^\mu\Psi_2 - \Psi_2D^\mu\Psi_1)]}. \tag{9}$$

The $-2p$ coefficient needs both gravity and scalar variations. Equation (9) also agrees with the local constrained kinetic action. Neither formula is guessed from a master-variable norm.

Put $\mathcal{D} = \Box_\gamma + 4$. The reconstruction $X = -3P'/(\rho^2 p)$ and (7) convert (8) to

$$\boxed{\omega^y = 9\sqrt{-\gamma}\alpha[(\mathcal{D}P_1)P_2' - (\mathcal{D}P_2)P_1']}. \tag{10}$$

For $P_i(y)Y_i(x)$ with $\Box Y_i = (\Lambda_i - 4)Y_i$, this is

$$\begin{aligned} \omega^y &= 9\sqrt{-\gamma}\alpha[\Lambda_1P_1P_2' - \Lambda_2P_1'P_2]Y_1Y_2, \\ \omega^\mu &= 9\sqrt{-\gamma}[\alpha P_1'P_2' + qP_1P_2](Y_1D^\mu Y_2 - Y_2D^\mu Y_1). \end{aligned} \tag{11}$$

Consequently the multiplier is 9Λ times the reduced Green current only at a common spectral value. A packet requires (10) or the mixed terms in (11). The independent identity

$$\partial_y\{\alpha[\Lambda_1P_1P_2' - \Lambda_2P_1'P_2]\} = (\Lambda_1 - \Lambda_2)(\alpha P_1'P_2' + qP_1P_2) \tag{12}$$

follows from (4) and cancels the tangential divergence in (11). A common-Λ replacement for unequal profiles fails this check.

3. Doubled-shell balance

Varying the common induced fields in (1) gives

$$\sum_i (K_{\mu\nu}^{(i)} - K_i h_{\mu\nu}) + \sigma h_{\mu\nu} = 0, \qquad \sum_i n_i\cdot\nabla\phi + \sigma_{,\phi} = 0. \tag{13}$$

Two pure-even or two pure-odd perturbations with the same one-side amplitude double (8)–(11); cross-parity bulk pairings cancel. In the generic even sector the shell is straight, and its condition in (4) cancels $\omega_+^y(T_b) + \omega_-^y(T_b)$ even for unequal Λ_i. There is no extra tangential kinetic shell current on this slice. The endpoint mass in the even reduced norm is not an independent physical shell kinetic field in this gauge.

The odd shell moves in longitudinal gauge. The complete moving-boundary action in [1] retains graph GHY minus slice GHY, the Einstein tangential divergence, the shifted bulk integration domain, and one common shell volume. In a graph $y = T_b + \zeta(x)$, the first two terms contribute $+3a_b\rho_b^2(D\zeta)^2$ per copy, while the shifted domain contributes $+3\rho_b^2 D\zeta\cdot D\psi_b$ per copy. The single shell contributes $-\sigma_b\rho_b^2(D\zeta)^2/2$. Using odd continuity $\psi_b = -a_b\zeta$, $X_b = -p_b\zeta$ and (3), their sum is $-3a_b\rho_b^2(D\zeta)^2$.

With $\zeta = -P_b/(a_b\rho_b^2)$, this fixes the completed kinetic action and its current:

$$S^{(2)}_{\Sigma,\mathrm{kin}} = -9c_o \int \sqrt{-\gamma}(DP_b)^2, \qquad \omega^\mu_\Sigma = 18c_o\sqrt{-\gamma}(P_{1b}D^\mu P_{2b} - P_{2b}D^\mu P_{1b}). \tag{14}$$

It includes both gravitational copies and one shell, rather than two shell tensions. The odd condition in (4) gives the exact balance

$$\partial_\mu \omega^\mu_\Sigma = \omega^y_+(T_b) + \omega^y_-(T_b). \tag{15}$$

Indeed both sides equal $18c_o\sqrt{-\gamma}(\Lambda_2 - \Lambda_1)P_{1b}P_{2b}Y_1Y_2$ for separated pairs. Thus

$$\partial_\mu \left[\int_\epsilon^{T_b} (\omega^\mu_+ + \omega^\mu_-)dy + \omega^\mu_\Sigma \right] = \omega^y_+(\epsilon) + \omega^y_-(\epsilon). \tag{16}$$

This is the conserved bulk-plus-shell current, subject to the lower-boundary flux. It is proved on the constrained generic class, not for arbitrary unreduced embeddings or additional boundary degrees of freedom.

At a common spectral root the integrated kinetic coefficients reproduce $K_e = 18\Lambda N_e$ and $K_o = 18\Lambda N_o$ from [1]. Their healthy sign does not make (8), (14) or (16) positive bilinears: presymplectic currents are antisymmetric and vanish when their two arguments coincide.

4. Vertex geometry and the radial endpoint

Use global dS_4 coordinates, $\gamma = -dt^2 + \cosh^2 t\, d\Omega_3^2$. The exact conformal radius obeys

$$\frac{dr}{r} = \frac{dy}{\rho}, \qquad r/y \to 1, \qquad \rho = r\Omega(r^2), \qquad dy = \Omega(r^2)dr.$$

With ambient coordinates $\mathsf{T} = r\sinh t$, $\mathsf{R} = r\cosh t$ and $u = re^{-t}$, $v = re^t$, the metric is exactly

$$g = \Omega(uv)^2 \left[du\, dv + \frac{(u+v)^2}{4} d\Omega_3^2 \right], \quad \mathsf{T} = (v-u)/2, \quad \mathsf{R} = (v+u)/2. \tag{17}$$

Here $\Omega(0) = 1$. Smooth even regular expansions give local smooth background extensions; they do not prescribe a global continuation. The radial chart covers the spacelike exterior of the cone. At fixed finite t, $y \to 0$ reaches the vertex. Reaching $u = 0$, $v > 0$ requires $t = \log(v/r) \to +\infty$. These are different limits, as already emphasized in the de Sitter-brane geometry of [6].

For smooth perturbations with bounded Cartesian first derivatives, the geometric current (5) is bounded near the vertex. The proper volume of $y = \epsilon$ over a fixed compact time interval is $O(\epsilon^4)$. Its flux therefore vanishes. This sufficient statement is independent of the divided master variable and does not control a time interval translated to infinity.

For the radial operator set

$$x = \int_y^{T_b} \frac{ds}{\rho(s)}, \quad F = \frac{P}{\rho^{3/2}|p|}, \quad s_L = \rho\left(\frac{3}{2}a + \frac{p'}{p}\right).$$

The potential is $V_L = s_{L,x} + s_L^2 + 2\rho^2 p^2/3 = 25/4 + O(e^{-2x})$, and

$$\alpha(P_1P_2' - P_1'P_2) = -W_x(F_1, F_2). \tag{18}$$

For the operator domain, $F \in H^2$ near infinity, so both F and F_x tend to zero. This proves the radial Green-form limit. The form domain H^1 alone gives no pointwise endpoint trace for F_x. Smooth disjoint

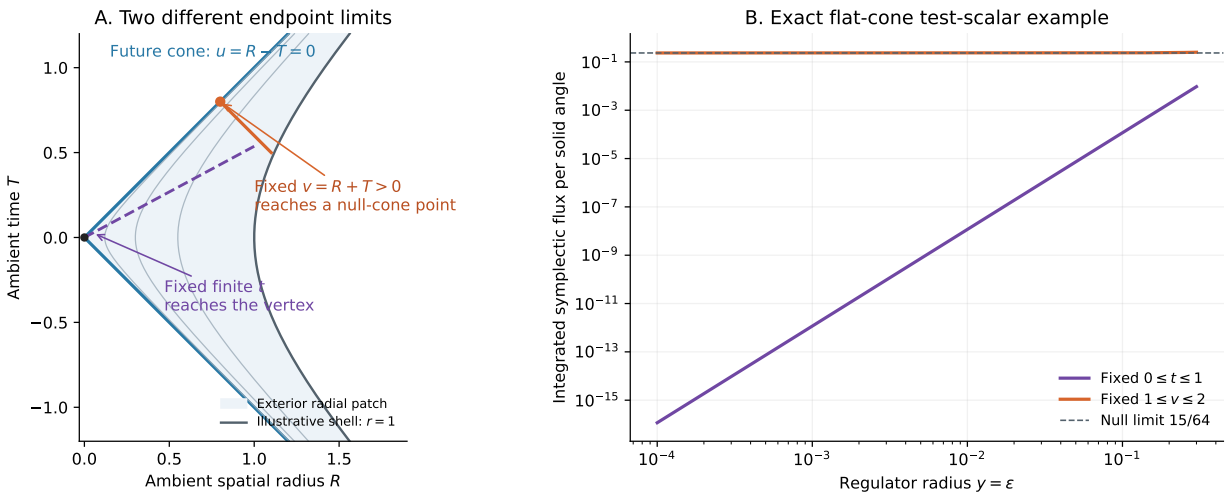


Figure 1: Distinct cone limits. The geometric panel uses conformal radius r and ambient (T, R). Any numerical flux panel refers only to the exact flat test-scalar example, not a registered gravitational wave packet.

bumps can give two H^1 functions whose Wronskian is one at an unbounded sequence and zero between supports; the closed variational identity does not assert a pointwise limit for all form vectors.

For the physical packet current (10), let $G = (\mathcal{D}P)/(\rho^{3/2}|p|)$. Then

$$\frac{\omega^y}{9\sqrt{-\gamma}} = -G_1 F_{2x} + G_2 F_{1x} + s_L(G_1 F_2 - G_2 F_1). \tag{19}$$

A sufficient strong finite-time condition is continuity on a compact time interval into $H^2_x(L^2(S^3))$ for F_i and into $H^1_x(L^2(S^3))$ for G_i, with enough angular regularity to define the current. Compactness of these time-indexed sets yields uniform tail control and zero integrated limit. No bound here is uniform over time intervals tending to infinity. Weak energy-space flux requires a separate trace or approximation construction.

5. Characteristic flux: a conditional result and an exact comparator

Conditional registered-model statement. Let two actual generic linearized solutions admit, uniformly on a compact segment $0 < v_1 \le v \le v_2$, the differentiated asymptotics

$$P_i = y^4 A_i(v, \vartheta) + O(y^6), \qquad \Psi_i = y^2 A_i + O(y^4), \qquad X_i = X_{0i} + O(y^2), \quad X_{0i} = -\frac{3}{\kappa}(4A_i + v\partial_v A_i). \tag{20}$$

Here $v = re^t$ and $\vartheta \in S^3$; remainders are smooth in (u, v, ϑ) with the uniform derivatives entering the current and equations. A C^1 bound suffices for current limits, and the invoked field equations require corresponding C^2 control. The last relation follows from radial momentum reconstruction. Replacing r by y changes only controlled subleading orders on this segment.

These leading fields are compatible with smooth metric components at the finite null surface: since $\delta g = 2\Psi g - 6\Psi dy^2$, their leading expression is

$$\delta g = 2uvA\, g - \frac{3}{2}A(v\, du + u\, dv)^2 + \text{smooth subleading tensor coefficients.} \tag{21}$$

This does not prove existence, a smooth global longitudinal representative, or smoothness at $u = v = 0$. An $O(u^2)$ remainder in Ψ can contribute $O(u)$ to the du^2 coefficient and must not be mislabeled $O(u^2)$ as a tensor remainder.

Directly applying (8) gives, per interior,

$$\omega^y\, dt \longrightarrow \frac{v^3}{8}\sqrt{\gamma_{S^3}}(X_{01}\partial_v X_{02} - X_{02}\partial_v X_{01})\, dv. \tag{22}$$

Indeed $\rho^4 \cosh^3 t = (v^3/8)y + O(y^3)$ and $X_i'|_t = (v/y)\partial_v X_{0i} + O(y)$; the scalar–metric term vanishes in this limit. The density transformation verifies that this is the actual null pullback. Its exact Jacobian in (17) gives

$$\omega^u = \frac{\Omega}{2r}\left(\frac{u}{\rho}\omega^y - u\omega^t\right) \longrightarrow \frac{v^3}{8}\sqrt{\gamma_{S^3}}(X_{01}\partial_v X_{02} - X_{02}\partial_v X_{01}). \tag{23}$$

Both components are necessary: omitting ω^t loses a factor of two. Equivalently $\sqrt{-g}g^{uv} \to v^3\sqrt{\gamma_{S^3}}/8$. The doubled parity flux in the $+y$ convention is consequently

$$\boxed{\lim_{y\to 0}\int \omega^y_{\text{doubled}} dt\, d\vartheta = \frac{1}{4}\int_{v_1}^{v_2} v^3 dv \int_{S^3} d\Omega_3 (X_{01}\partial_v X_{02} - X_{02}\partial_v X_{01})}. \tag{24}$$

The outward flux from the exterior has the opposite sign. This is an antisymmetric symplectic flux, not a positive energy flux.

The leading bilinear is not identically zero: choosing $X_{01} = f(v)H(\vartheta)$, $X_{02} = vf(v)H(\vartheta)$ gives Wronskian f^2H^2. Equation (20) can be solved for the corresponding leading A_i on the positive segment. These are candidate leading data only. No pair of global registered solutions with these data, the distant shell conditions and the required gauge regularity has been constructed here.

Unconditional flat comparator. A massless test scalar in five-dimensional Minkowski space has radial operator $-y^{-2}(y^4 f')'$, Hilbert weight y^2, and form $\int y^4 |f'|^2 dy$. At zero mass its radial solutions are 1 and y^{-3}; only the first is square integrable at zero. The endpoint is limit-point, with Liouville potential 9/4. Consider

$$\varphi_1 = 1, \qquad \varphi_2 = \mathsf{T} = y \sinh t. \tag{25}$$

These are smooth local exact solutions. To realize them with finite ambient energy, choose compact smooth initial data at $\mathsf{T} = 0$ equal to $(\varphi_1, \dot\varphi_1) = (1, 0)$ and $(\varphi_2, \dot\varphi_2) = (0, 1)$ on a ball of radius R_0. Finite propagation speed gives (25) wherever $|\mathsf{T}| + \mathsf{R} < R_0$. Thus any cone segment $0 < a \le v \le b < R_0$ has a neighborhood where the finite-energy solutions equal (25). A remote boundary outside that causal diamond does not affect the calculation.

At each finite t the Liouville profiles are $y^{3/2}$ and $y^{5/2}\sinh t$, both H^2 near the endpoint. Their current per solid angle on $y = \epsilon$ obeys

$$\int_{t_1}^{t_2} j^y dt = \frac{\epsilon^4}{4}[\cosh^4 t_2 - \cosh^4 t_1] \longrightarrow 0, \tag{26}$$

whereas on $t \in [\log(a/\epsilon), \log(b/\epsilon)]$,

$$j^y dt = \frac{(v + \epsilon^2/v)^3(v - \epsilon^2/v)}{16v} dv \longrightarrow \frac{v^3}{16} dv, \qquad \int j^y dt \longrightarrow \frac{b^4 - a^4}{64} \neq 0. \tag{27}$$

The convergence is uniform since $a > 0$. The second field separately has nonzero null derivative and a nonzero quadratic stress-energy component; this is distinct from the antisymmetric pairing in (27).

The example disproves an inference from radial limit-point closure, smooth extension and finite energy to zero characteristic flux. It is a test-field comparator, not a solution of the registered coupled Einstein–scalar shell problem. It contradicts no self-adjoint radial theorem. Free fields on de Sitter leaves already admit established ambient spectral decompositions under additional state and convergence hypotheses [7].

Energy-completion comparator. The same exact flat test field also distinguishes two energy topologies. On $\mathsf{T} = t = 0$, take zero velocity and $f(y) = \chi_{\rm out}(y) y^{-\nu}$, where the outer cutoff is smooth and equals one near zero. For $1 \le \nu < 3/2$,

$$N[f] \sim \int_0 y^{2-2\nu} dy < \infty, \qquad Q[f] \sim \nu^2 \int_0 y^{2-2\nu} dy < \infty, \qquad E_{\mathsf{T},\rm grad}[f] \sim \nu^2 \int_0 y^{1-2\nu} dy = \infty. \quad (27a)$$

For $\nu = 5/4$ the powers are $-1/2$ and $-3/2$, respectively. Inner smooth cutoffs converge in $N + Q$, with cost $O(\epsilon^{3-2\nu})$, while their inertial gradient energies diverge. The different weights reflect the lapse y of the de Sitter time clock: its radial gradient contribution is $\int y^4 |f'|^2 dy$, whereas an inertial slice has $\int y^3 |f'|^2 dy$. This example uses a singular energy-limit datum, in contrast to the smooth finite-energy solutions in (25)–(27). It disproves a blanket identification of these two test-field completions. It establishes no equivalence or inequivalence of registered gravitational phase spaces. Specifying which energy is completed is indispensable.

6. Tensor regularity and energy completion

The even tensor zero profile $h_T(y) = 1$ has norm $\int_0^{T_b} \rho^2 dy$. Let $q_{\mu\nu}$ be a slice TT Einstein graviton, $\delta R_{\mu\nu}[\gamma; q] = 3 q_{\mu\nu}$. Its lift is $\delta g_{\mu\nu} = \rho^2 q_{\mu\nu}$, $\delta\phi = 0$. For any four-dimensional Einstein slice, warped curvature gives

$$C^{(5)}_{\mu\nu\alpha\beta} = \rho^2 C^{(4)}_{\mu\nu\alpha\beta}, \qquad C^{(5)}_{y\mu y\nu} = C^{(5)}_{y\mu\nu\alpha} = 0.$$

Linearization and an orthonormal frame therefore give

$$\delta C^{(5)}_{\hat a\hat b\hat c\hat d} = \rho^{-2} \delta C^{(4)}_{\hat a\hat b\hat c\hat d}. \quad (28)$$

The background Weyl tensor vanishes, so this perturbation is gauge invariant. Along a finite-t ray the frame has a bounded Cartesian limit. Nonzero slice Weyl implies a y^{-2} divergence and excludes a C^2 metric extension through the vertex. Tensor components, rather than the Lorentzian contraction C^2, are needed because nonzero radiative Weyl tensors can have vanishing scalar contractions. Directions with zero slice Weyl are not excluded by this argument.

This diagnostic is established prior art: Garriga and Sasaki [6, section III, p. 9] display $C_5^2 = C_4^2/a^4$ and explain the singular normalizable zero mode as part of a mode expansion. Equation (28) applies the mechanism componentwise to the nonconstant scalar warp; it is not a novel discovery.

The smoothness obstruction does not eliminate the mode from an energy completion. For a smooth cutoff $h_\epsilon = \chi(y/\epsilon)$, zero below ϵ and one above 2ϵ, the Neumann shell condition is unchanged and

$$\int \rho^2 |h_\epsilon - 1|^2 dy = O(\epsilon^3), \qquad \int \rho^4 |h_\epsilon'|^2 dy = O(\epsilon^3). \quad (29)$$

These are initial-data approximants, not separated zero-mass eigenfunctions. For global S^3 tensor harmonics, data on the regular slice $\mathsf{T} = 0$ have bounded Cartesian metric components, with derivatives and velocities $O(r^{-1})$. Their local $H^1 \times L^2$ energy tail is $\int_0^\epsilon r^3 r^{-2} dr = O(\epsilon^2)$; the cutoff derivative costs the same order. Radial cutoffs preserve angular TT metric and extrinsic-curvature data. Their full spatial divergence remains zero because its radial term is proportional to the angular trace. On the

time-symmetric background slice the angular Ricci tensor is isotropic, the scalar gradient is radial, and $\delta\phi = \delta\pi_\phi = 0$; the linearized Hamiltonian and momentum constraints are preserved.

Thus pointwise smooth configurations and finite-energy limits differ here. No deletion of massless gravity, inequivalence of completed physical spaces, or global evolution theorem follows from (28)–(29).

7. Gauge charges and exceptional subspaces

For a field-independent vector ξ, the on-shell covariant identity is

$$\boldsymbol{\omega}(\delta, \mathcal{L}_\xi) = d\boldsymbol{k}_\xi(\delta), \quad \boldsymbol{k}_\xi = \delta\boldsymbol{Q}_\xi - i_\xi\boldsymbol{\theta}, \quad \boldsymbol{Q}_\xi = -\tfrac{1}{2} \star d\xi^\flat. \tag{30}$$

Field-dependent vectors require the $-Q_{\delta\xi}$ correction. Smooth Cartesian fields, perturbations and vector generators with bounded required derivatives have bounded k_ξ. A cut $y = \epsilon, t = t_0$ has area $O(\epsilon^3)$ at fixed finite t_0, hence

$$\int_{S^3_{\epsilon,t_0}} k_\xi = O(\epsilon^3) \to 0. \tag{31}$$

This sufficient vertex-charge statement gives no result on a finite-radius null cut. At the shell the embedding must transform with the bulk fields. Generators supported away from physical boundaries are degeneracies; other generators require the total boundary pairing in the specified phase space.

For precision, the Hessian-degenerate subspace is $D_\mu D_\nu Y = -\gamma_{\mu\nu} Y$, locally the five embedding harmonics $Y = c_A n^A$ with $X^A = r n^A$. Its Gaussian-normal residual is $\xi^y = e_0 Y$, $\xi^\mu = (\ell_0 - e_0 F) D^\mu Y$, $F' = \rho^{-2}$. Fix the additive constant by the regular odd Laurent expansion. With $s = r^2$, $A = -rF$ and $B = (\Omega^{-1} - A)/s$, its translation part is

$$\xi^A = e_0[A(s)c^A + B(s)(c_B X^B)X^A], \qquad A - 2sA_s = \Omega^{-1}, \quad A(0) = 1. \tag{32}$$

Both coefficients are smooth. For example $F = -1/r + \int_0^r [\Omega(v^2)^{-1} - 1]v^{-2}dv$ exhibits this explicitly. The apparent $1/y$ tangential coefficient represents a regular Cartesian translation. In contrast, the remaining vector is $\ell_0[rc^A - (c_B X^B)X^A/r]$, generically not continuous in the full ambient approach to the vertex, and hence not C^1. For example, with $c = (0, 1, 0, \ldots)$, $X^1 = \epsilon$ and $X^0 = \sqrt{\epsilon^2 - \epsilon^6}$, one has $r = \epsilon^3$ and $\xi^1 = \ell_0(\epsilon^3 - \epsilon^{-1})$. Bounded finite-$t$ rays do not detect this divergence. A smooth gauge convention fixes that constant; no charged physical mode is inferred from its exclusion. A weaker completed gauge group needs its own pairing analysis.

The background equation also gives $a + F = -\frac{1}{3}\int_0^y p(v)^2 dv = O(y^3)$. The earlier relative-displacement exclusion uses $p_b B_b \neq 0$ and simultaneous shell matching. The constant harmonic instead uses the static Jacobi determinant in [1]. Neither argument permits dropping all harmonics at $\mu^2 = 0$ or $\mu^2 = -4$; other harmonics at these values remain generic. We prove no global exceptional-charge classification or global longitudinal-gauge coverage.

8. Interpretation, prior art and reproducibility

Equations (8)–(16) supply an explicit action-derived current and shell balance for this constrained solution class, including unequal spectral values. Equation (24) is a conditional identity for actual solutions with the stated uniform asymptotics. Equation (27) is an unconditional counterexample to a general zero-characteristic-flux inference. Equations (28)–(32) distinguish smoothness, completion and local gauge charge. All leave the earlier radial positivity theorem intact.

The underlying methods and several diagnostics predate this calculation. Iyer–Wald [2] describes the potential and charge ambiguities; Wald–Zoupas [3] requires a boundary field space and suitable

extension of the full current for radiative charges. Harlow–Wu [4] treats variational and corner terms, explicitly excluding null boundaries from its main construction. Null charges in [5] concern a specified four-dimensional vacuum field space, not this expanding five-dimensional matter cone and shell. Garriga–Sasaki [6] supplies the direct horizon and zero-mode precedent. Bertola et al. [7] give an ambient spectral dictionary under additional hypotheses. Gen–Sasaki [8] treats displacement and moving-boundary terms, while Kofman–Martin–Peloso [9] derives physical normalization including boundary terms.

The recent null-hypersurface analysis of Ciambelli–Freidel–Leigh [10] was also inspected. Its selected sections explicitly retain boundary-kernel, corner and function-space choices; its four-dimensional vacuum kinematical reduction is not a completed constrained phase space for this model. The accompanying literature report records bounded queries and primary passages read. It establishes neither exhaustive coverage nor priority for this exact application. The claim is the displayed model-specific calculation, not a new phase-space method, threshold, or physical mechanism.

The four accompanying derivations and their exact checkers verify the potential curl, mixed-spectral conservation, shell signs, Liouville conversion, density Jacobian, comparator flux and tensor/gauge identities. Deliberately wrong formulas and optimization-mode runs test the checkers. Independent internal calculations agree on the current. These receipts support algebra and exact scaling calculations; they are not formal proofs of functional analysis, hyperbolic existence or completeness.

The next task is to specify a regular Cauchy-data completion or a characteristic boundary-data space, prove existence and shell compatibility, control the reconstructed current under that completion, and identify its zero-charge gauge quotient. Compactly supported tangential test variations used in local action integrations are not arbitrary nonzero on-shell solutions compactly supported in time. Finite-slab evolution keeps its Cauchy terms, and the global S^3 sections avoid unmentioned spatial flux. No full registered radiative packet, global stability theorem, branch transition, reheating process, or evidence for a higher-dimensional cosmological origin is established here.

Source and data availability. The prior checkpoint is DOI 10.5281/zenodo.23335485, *HDBLAST: Conditional Scalar Spectral Positivity and Tensor Mass Thresholds on Regular de Sitter Shells*, with immutable source 401efdfaaef657bcbdaec588b21b82789daddd0e. This continuation and its technical notes, checkers and review records are distributed in the accompanying GitHub capsule. There is no separate DOI for this continuation. DOI 10.5281/zenodo.23335485 identifies the prior spectral checkpoint only; that edition and its immutable source remain unchanged.

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